Sagarika Manjunath Chavan

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SUMMARY

AI/ML Engineer with expertise in Data Science, Data Engineering, Cybersecurity, Full-Stack Development, and AI Research with **3+ years** of experience building production-scale machine learning systems, LLM-powered applications, and AI-driven cybersecurity solutions across enterprise environments. Experienced in developing intelligent threat detection, network traffic classification, and SOC automation pipelines processing millions of security events and encrypted network flows using cloud-native ML technologies on AWS and Azure. Currently pursuing a Master of Computer Science (AI, Data Science & Cybersecurity) at the University of Sydney, researching graph-aware reinforcement learning techniques to enhance autonomous LLM agents for adaptive penetration testing and intelligent cyber defence.

EXPERIENCE

Data Engineer · **Fiorin, Sydney, Australia** Jul 2026 – Present

- Building a **data engineering pipeline** to ingest, map, reconcile, and validate construction budget data, transforming raw Excel exports into structured cost records and implementing **automated over-budget flagging** to identify cost variances across **$2.2B+ of project data**.

Data Analyst Intern · **Canopus Networks, Sydney, Australia** Jan 2026 – Apr 2026

- Developed an **LLM-powered classification pipeline** for **3M+ encrypted network flows** using SNI and domain metadata, generating structured JSON outputs with confidence scores and reducing false positives by **18%.**
- Analysed network flow features (uplink/downlink ratio, bytes-per-flow, and session duration) to classify enterprise research traffic, and developed the **SuperNetFlow** web application for traffic analysis and visualisation.

Data Analytics & ML Intern · **Cubic Innov8, Sydney, Australia** Mar 2025 – Jul 2025

- Developed an **AI-powered SOC remediation pipeline** using **XGBoost, Random Forest, and CatBoost** to classify **9.5M+ Microsoft GUIDE security alerts**. Leveraged TF-IDF and NLP to extract threat intelligence from incident descriptions, reducing analyst triage time by **10%** through false-positive analysis and threshold tuning.

Senior Software Engineer (Data Engineer) · **Bosch Global Software Technologies, Bangalore, India / Germany** Aug 2023 – Feb 2025

- Designed a logarithmic chunking algorithm enabling **10 GB ADAS video** streaming with **60% faster startup** and **95%+** marker–video synchronisation accuracy.
- Improved ML vector embedding compatibility by **30%** across **5+ models** using **Azure ML**, contributing to AI-driven ADAS workflows; recognised with **Certificates of Appreciation** at Bosch EDE and EDE2 department meetings.
- Resolved architectural challenges in an Angular/React micro-frontend using Redux and collaborated onsite in Germany with cross-functional teams on the ADAS PACE platform.

Software Engineering Intern · **Bosch Global Software Technologies, Bangalore, India** Jan 2023 – May 2023; Jun 2022 – July 2022

- Automated an MLOps pipeline for traffic image anonymisation using Docker and cloud technologies, and developed geospatial visualisations with S2 geometry and Supercluster.

EDUCATION

Masters in Computer Science - AI, Data Science & Cybersecurity · **University of Sydney, Sydney, Australia** Feb 2025 – Dec 2026

WAM: 81.1 · Sydney International Student Award (20% tuition scholarship)

Research Topic: ***Enhancing LLM Agents for Automated Penetration Testing using Reinforcement Learning***

B.Tech Computer Science - Machine Intelligence & Data Science · **PES University, Bangalore, India** Aug 2019 – Sep 2023

GPA: 8.47/10 · Distinction Award Recipient · MRD Merit Scholarship (Top 20% of Cohort)

PROJECTS

- **Autonomous LLM Agents for Penetration Testing (Research)** *(Python, Graph Neural Networks, Reinforcement Learning, LLMs)* - Researching graph-aware RL for autonomous LLM agents that use penetration graphs to plan and adapt multi-stage penetration testing.
- **Real-Time Electricity Grid Data Platform** *(Python, MQTT, Plotly Dash)* - Designed an end-to-end real-time data engineering pipeline using MQTT and Plotly Dash for live ingestion and geospatial visualisation of Australian electricity grid data.
- **Cancer Detection using Deep Learning** *(Python, TensorFlow, Keras, CNN)* - Developed a CNN-based histopathologic image classifier achieving **92% accuracy** using batch normalisation and dropout.

PUBLICATIONS

Automated Script Evaluation using Machine Learning & Natural Language Processing — IEEE INOCON 2023 & IJISAE 2023

Designed an AI-powered automated answer evaluation system using **BERT, GPT-3, and Google Cloud OCR**, achieving **90%+ scoring accuracy** and reducing evaluation time by **80%**.

TECHNICAL SKILLS

Languages	Python, Java, C, JavaScript, TypeScript, SQL
AI / ML	Machine Learning, Deep Learning, LLMs, Reinforcement Learning, Graph Neural Networks (GNNs), Prompt Engineering, RAG, NLP, PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, OpenCV
Data Science / Engineering	Pandas, NumPy, SciPy, Matplotlib, Plotly, Apache Spark, Hive, Kafka, MQTT, Hadoop, Azure Databricks, ETL Pipelines, Real-Time Data Streaming, MySQL, MongoDB, Cosmos DB
Cloud / MLOps	AWS, Azure ML, Azure DevOps, Docker, Kubernetes, Git, GitHub Actions, CI/CD
Cybersecurity	Penetration Testing, Mobile Reverse Engineering, Burp Suite, Wireshark, Nmap, Frida, Kali Linux
Full Stack Development	React, Angular, Node.js, Flask, FastAPI, GraphQL